

TASK 9: Collections Framework – Student Management System

Tools:

- Primary: IntelliJ IDEA
- Alternatives: Eclipse

Hints / Mini Guide:

1. Store student objects in ArrayList.
2. Use HashMap for fast lookup.
3. Implement sorting using Comparator.
4. Remove duplicates using Set.
5. Iterate using enhanced for loop.
6. Demonstrate collection hierarchy.
7. Optimize memory usage.
8. Print formatted reports.

Deliverables:

- Student management console app
- Sorted and filtered outputs

Interview Questions Related To Above Task:

- List vs Set vs Map?
- Why HashMap is faster?
- What is Comparator?
- Internal working of HashMap?
- When to use LinkedList?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📌 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

